

# AI-Enabled Quality Engineering and Application Security

## Executive discovery questionnaire - role-routed, multiple-selection version (v2)

**Purpose:** tailor the EVP presentation, select relevant industry research, identify practical QA and application-security opportunities, and define a conservative phased pilot. Approximate ranges are sufficient; no customer data, source code, detailed rate cards, or formal financial commitments are required.

### How to complete (about 15 minutes for your route)

Answer only the sections for your role and leave the others blank:

Executive sponsor, Finance, procurement: Sections 1, 5B and 6.

Engineering, delivery or QE leader: Sections 2, 3, 4 and 6.

Application security leader: Sections 4 (Q18-Q20), 5A and 6.

Square boxes allow multiple selections; respect the limit shown. Dropdown fields take one answer; in the grid question (Q17) tick one box per row. Click fields to enter text, then save the PDF normally.

### Respondent information

Your role in this discussion (routes the questionnaire)

Name

Title / role

Function / business area

Date

### 1. Executive objectives and spending concerns

Executive sponsor and Finance route. Establish what the EVP expects the conversation to solve and how success should be described.

#### 1. What are the primary objectives for exploring AI in quality engineering?

Select up to three.

- |                                                                                  |                                                                |
|----------------------------------------------------------------------------------|----------------------------------------------------------------|
| <input type="checkbox"/> Reduce manual testing effort                            | <input type="checkbox"/> Shorten regression and release cycles |
| <input type="checkbox"/> Improve automation coverage and maintainability         | <input type="checkbox"/> Reduce escaped production defects     |
| <input type="checkbox"/> Improve application-security remediation                | <input type="checkbox"/> Improve audit and release evidence    |
| <input type="checkbox"/> Increase delivery capacity without increasing team size | <input type="checkbox"/> Improve developer productivity        |

#### 2. If a pilot proved one thing, which ONE outcome would justify continuing?

Select one. This defines what "success" will mean for the executive sponsor.

#### 3. Which outcomes must NOT deteriorate for a pilot to count as a success?

Select all that apply. These become the quality and security floors.

- |                                                                  |                                                                   |
|------------------------------------------------------------------|-------------------------------------------------------------------|
| <input type="checkbox"/> Escaped defects or production incidents | <input type="checkbox"/> Release stability or change-failure rate |
| <input type="checkbox"/> Security posture and remediation SLAs   | <input type="checkbox"/> Privacy and sensitive-data handling      |
| <input type="checkbox"/> Audit and release evidence              | <input type="checkbox"/> Engineer adoption and morale             |

## 1. Executive objectives and spending concerns

Continued

### 4. What are the greatest concerns with current or previous AI initiatives?

Select up to three.

- |                                                                               |                                                                                      |
|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <input type="checkbox"/> Unclear or unproven business value                   | <input type="checkbox"/> Excessive consulting or implementation cost                 |
| <input type="checkbox"/> High licensing, token, or model-consumption cost     | <input type="checkbox"/> Too many overlapping tools or platforms                     |
| <input type="checkbox"/> Low user adoption or weak workflow fit               | <input type="checkbox"/> Security, privacy, or data-residency risk                   |
| <input type="checkbox"/> Difficulty moving beyond pilots                      | <input type="checkbox"/> Weak governance, controls, or accountability                |
| <input type="checkbox"/> Productivity gains that do not become budget savings | <input type="checkbox"/> Quality or security regressions from AI-generated artifacts |
| <input type="checkbox"/> Regulatory or audit scrutiny of AI use               | <input type="checkbox"/> Previous AI pilots that did not deliver                     |

## 2. Delivery scope and QA operating model

Engineering, delivery and QE route. Approximate ranges are sufficient; detailed rate cards and organization charts are not required.

### 5. Approximately how many delivery teams or applications are in the potential scope?

Select one range.

Target business area, portfolio, or application type (optional):

### 6. Roughly what share of in-scope applications are modern cloud or API-based systems with automated build and deployment pipelines?

Select one range.

### 7. For a typical delivery squad, provide the approximate number of resources by role.

Enter approximate numbers for a typical or median squad; ranges are acceptable (for example 6 / 2 / 1 / 1 / shared).

| Role / capability                                 | Approx. number       |
|---------------------------------------------------|----------------------|
| Developers                                        | <input type="text"/> |
| Manual QA analysts                                | <input type="text"/> |
| Automation engineers / SDETs / QEs                | <input type="text"/> |
| Business analysts and product owners              | <input type="text"/> |
| AppSec or security resources (embedded or shared) | <input type="text"/> |

### 8. Approximately what percentage of QA headcount is external, offshore, or delivered through a managed service?

Select one range (by headcount, not spend).

## 2. Delivery scope and QA operating model

Continued

### 9. Which descriptions apply to the current quality-engineering operating model?

Select all that apply.

- |                                                                      |                                                                       |
|----------------------------------------------------------------------|-----------------------------------------------------------------------|
| <input type="checkbox"/> Centralized QA organization                 | <input type="checkbox"/> QA / QE embedded in delivery squads          |
| <input type="checkbox"/> Managed-service model                       | <input type="checkbox"/> Hybrid centralized and embedded model        |
| <input type="checkbox"/> Dedicated automation / SDET capability      | <input type="checkbox"/> Business-led UAT outside the QA organization |
| <input type="checkbox"/> Model varies significantly by business unit | <input type="checkbox"/> Operating model is being redesigned          |

### 10. Where does the QA organization spend the most human effort?

Select your top THREE.

- |                                                                                 |                                                                                 |
|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <input type="checkbox"/> Requirements review and acceptance-criteria refinement | <input type="checkbox"/> Test planning, scenario design, and test-case creation |
| <input type="checkbox"/> Building automated tests                               | <input type="checkbox"/> Maintaining or repairing automated tests               |
| <input type="checkbox"/> Preparing test data                                    | <input type="checkbox"/> Preparing or troubleshooting test environments         |
| <input type="checkbox"/> Manual functional test execution                       | <input type="checkbox"/> Regression execution                                   |
| <input type="checkbox"/> Failed-test triage and root-cause investigation        | <input type="checkbox"/> Defect creation, routing, retesting, and closure       |
| <input type="checkbox"/> UAT coordination and support                           | <input type="checkbox"/> Quality reporting, traceability, and release evidence  |

## 3. Testing maturity and workflow

Engineering, delivery and QE route. Identify where human effort, wait time, rework, and quality risk are concentrated today.

### 11. Roughly what share of regression test cases execute automatically without human intervention?

Select one range.

### 12. Which test types are consistently automated within CI/CD?

Select all that apply.

- |                                     |                                      |                                                              |
|-------------------------------------|--------------------------------------|--------------------------------------------------------------|
| <input type="checkbox"/> Unit       | <input type="checkbox"/> Component   | <input type="checkbox"/> API                                 |
| <input type="checkbox"/> Contract   | <input type="checkbox"/> Integration | <input type="checkbox"/> UI / end-to-end                     |
| <input type="checkbox"/> Regression | <input type="checkbox"/> Performance | <input type="checkbox"/> Accessibility                       |
| <input type="checkbox"/> Security   | <input type="checkbox"/> Few or none | <input type="checkbox"/> Varies significantly by application |

### 13. What is the typical elapsed regression-testing duration for a major release?

Select one range.

### 14. Approximately how many active human hours does a full regression cycle require?

Select one range. Count hands-on time across all people, not elapsed time.

### 3. Testing maturity and workflow

Continued

#### 15. Which issues most frequently delay testing or releases?

Select up to three.

- |                                                                       |                                                                          |
|-----------------------------------------------------------------------|--------------------------------------------------------------------------|
| <input type="checkbox"/> Incomplete or changing requirements          | <input type="checkbox"/> Insufficient unit or component testing          |
| <input type="checkbox"/> Manual test-case creation                    | <input type="checkbox"/> Manual regression execution                     |
| <input type="checkbox"/> Unstable automation or flaky tests           | <input type="checkbox"/> Test-data availability                          |
| <input type="checkbox"/> Test-environment availability or instability | <input type="checkbox"/> Downstream-system dependencies                  |
| <input type="checkbox"/> Legacy or mainframe integration              | <input type="checkbox"/> Defect triage and ownership                     |
| <input type="checkbox"/> Security findings or remediation             | <input type="checkbox"/> UAT, release evidence, or approval requirements |

#### 16. Which quality and security checks are mandatory release gates?

Select all that apply.

- |                                                                       |                                                                    |
|-----------------------------------------------------------------------|--------------------------------------------------------------------|
| <input type="checkbox"/> Unit-test pass rate and/or code coverage     | <input type="checkbox"/> API, integration, or regression pass rate |
| <input type="checkbox"/> Critical and high-priority defect thresholds | <input type="checkbox"/> SAST findings                             |
| <input type="checkbox"/> Software composition analysis / CVE findings | <input type="checkbox"/> DAST or penetration-testing findings      |
| <input type="checkbox"/> Performance or resilience thresholds         | <input type="checkbox"/> Accessibility compliance                  |
| <input type="checkbox"/> UAT or business approval                     | <input type="checkbox"/> Production-readiness / change approval    |
| <input type="checkbox"/> Traceability, audit, or control evidence     |                                                                    |

### 4. Metrics, toolchain, and AI readiness

Engineering, QE and AppSec routes. These responses determine whether an EVP-level savings hypothesis can be evidence-based.

#### 17. For each baseline measure, indicate whether it is tracked and trusted, tracked but unreliable, or not tracked.

Select one per row. Leave a row blank if you do not know.

| Measure                                                   | Tracked and trusted      | Tracked but unreliable   | Not tracked              |
|-----------------------------------------------------------|--------------------------|--------------------------|--------------------------|
| QA effort hours by activity, release, or application      | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Regression duration and execution volume                  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Automation coverage and maintenance effort                | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Flaky-test or rerun rate                                  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Defect volume, severity, reopen rate, and escaped defects | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Change-failure, rollback, or production-incident rate     | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Mean time to triage and remediate defects                 | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Release frequency and lead time for change                | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| QA labour, contractor, managed-service, or tool spend     | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Security finding backlog and remediation SLA performance  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

#### 4. Metrics, toolchain, and AI readiness

Continued

##### 18. Which enterprise AI platform is approved for engineering pilots, and which coding or testing assistants are approved for engineers today?

Select one platform; list assistants if known.

Approved enterprise AI platform

Approved coding or testing assistants (for example GitHub Copilot, test-tool AI features):

##### 19. Identify the primary tools currently used.

Optional. Product names are helpful but not required.

| Capability                     | Current platform or tool |
|--------------------------------|--------------------------|
| Requirements / work management | <input type="text"/>     |
| Source control and CI/CD       | <input type="text"/>     |
| Test management and automation | <input type="text"/>     |
| Observability / logs / traces  | <input type="text"/>     |
| SAST / SCA / DAST / AppSec     | <input type="text"/>     |

##### 20. Which AI-assisted QA or AppSec capabilities are already in use or being evaluated?

Tick "In use" and/or "Evaluating" for each row that applies.

|                                                                | In use                   | Evaluating               |
|----------------------------------------------------------------|--------------------------|--------------------------|
| Requirements, acceptance criteria, or test-scenario generation | <input type="checkbox"/> | <input type="checkbox"/> |
| Unit, API, or UI test generation                               | <input type="checkbox"/> | <input type="checkbox"/> |
| Test automation maintenance or self-healing                    | <input type="checkbox"/> | <input type="checkbox"/> |
| Regression-test selection or prioritization                    | <input type="checkbox"/> | <input type="checkbox"/> |
| Synthetic test-data generation                                 | <input type="checkbox"/> | <input type="checkbox"/> |
| Failed-test triage or root-cause analysis                      | <input type="checkbox"/> | <input type="checkbox"/> |
| Defect creation, classification, or routing                    | <input type="checkbox"/> | <input type="checkbox"/> |
| Release-quality summaries or evidence preparation              | <input type="checkbox"/> | <input type="checkbox"/> |
| Vulnerability triage, remediation guidance, or code fixes      | <input type="checkbox"/> | <input type="checkbox"/> |
| None / only informal experimentation                           | <input type="checkbox"/> | <input type="checkbox"/> |

#### 5A. Application security and AI controls

Application security route. Separate task-level productivity from the control boundaries a first pilot must respect.

##### 21. Which AI-assisted application-security use cases would be most valuable?

Select up to three.

- |                                                                               |                                                                                   |
|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| <input type="checkbox"/> Contextual SAST / SCA / DAST finding prioritization  | <input type="checkbox"/> False-positive, reachability, or exploitability analysis |
| <input type="checkbox"/> Developer-friendly vulnerability explanations        | <input type="checkbox"/> Secure-code remediation recommendations                  |
| <input type="checkbox"/> Remediation pull-request generation for human review | <input type="checkbox"/> Security-test and regression-test generation             |
| <input type="checkbox"/> AI-assisted threat modeling                          | <input type="checkbox"/> Fix validation and vulnerability-closure evidence        |

## 5A. Application security and AI controls

Continued

### 22. What is the HIGHEST level of AI action acceptable during an initial pilot?

Select one. Each level includes the levels above it.

### 23. Which AI controls are NOT yet in place for AI tooling and would need to be addressed before a pilot?

Select all that apply. Leave blank if all are in place.

- |                                                                                        |                                                                                            |
|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| <input type="checkbox"/> Source-code and intellectual-property confidentiality         | <input type="checkbox"/> Customer-data privacy and sensitive-data handling                 |
| <input type="checkbox"/> Data residency and retention                                  | <input type="checkbox"/> Approved-model and vendor enforcement                             |
| <input type="checkbox"/> Role-based access and segregation of duties for AI identities | <input type="checkbox"/> Audit logging and traceability of prompts, outputs, and approvals |
| <input type="checkbox"/> Human approval for production-impacting actions               | <input type="checkbox"/> Explainability and supporting evidence                            |
| <input type="checkbox"/> Model, prompt, and generated-artifact versioning              | <input type="checkbox"/> License, token, and model-consumption cost controls               |
| <input type="checkbox"/> Vendor portability and ability to disable AI safely           | <input type="checkbox"/> Model inventory and risk rating (OSFI E-23 readiness)             |

## 5B. Economics and financial capture

Executive sponsor and Finance route. A productivity gain becomes a saving only when Finance can name the budget line that changes.

### 24. Which forms of spending or capacity could realistically be reduced, avoided, or redeployed within the next 12 months?

Select all that apply.

- |                                                                                |                                                                          |
|--------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <input type="checkbox"/> Contractor renewals falling due in the next 12 months | <input type="checkbox"/> Managed-service scope or unit volumes           |
| <input type="checkbox"/> Planned QA or security hiring                         | <input type="checkbox"/> Overtime or surge capacity                      |
| <input type="checkbox"/> Testing or security-tool licenses                     | <input type="checkbox"/> Test infrastructure or execution cost           |
| <input type="checkbox"/> Defect rework and production-incident effort          | <input type="checkbox"/> Capacity redeployed to additional delivery work |
| <input type="checkbox"/> No capture mechanism has yet been identified          | <input type="checkbox"/> Not my decision                                 |

### 25. Approximately what is the annual addressable QA and testing spend in scope (internal labour plus external services and tooling)?

Select one range. Prefer not to say is acceptable.

### 26. Roughly what share of that spend is variable (contractors, offshore, managed services) rather than employees?

Select one range.

### 27. If QA capacity were released, what would most likely happen to it?

Select one.

## 5B. Economics and financial capture

Continued

### 28. What is the minimum evidence Finance would accept to recognize a saving?

Select one.

## 6. Pilot direction and executive input

All routes. A conservative pilot should be limited, measurable, and subject to an explicit go/no-go decision.

### 29. Which application types could be suitable for an initial pilot?

Select all that apply.

- |                                                                     |                                                                        |
|---------------------------------------------------------------------|------------------------------------------------------------------------|
| <input type="checkbox"/> Modern API or microservices application    | <input type="checkbox"/> Internal web application                      |
| <input type="checkbox"/> Customer-facing web application            | <input type="checkbox"/> Mobile application                            |
| <input type="checkbox"/> Legacy or mainframe-integrated application | <input type="checkbox"/> Data or analytics application                 |
| <input type="checkbox"/> Packaged platform                          | <input type="checkbox"/> More than one application type for comparison |
| <input type="checkbox"/> Candidate not yet selected                 |                                                                        |

### 30. Which use cases could be suitable for a limited pilot?

Select up to three.

- |                                                                                   |                                                                               |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| <input type="checkbox"/> Requirement analysis and test-scenario generation        | <input type="checkbox"/> Executable API or component-test generation          |
| <input type="checkbox"/> UI automation generation or maintenance                  | <input type="checkbox"/> Change-impact analysis and regression-test selection |
| <input type="checkbox"/> Failed-test triage and root-cause analysis               | <input type="checkbox"/> Flaky-test detection and automation repair           |
| <input type="checkbox"/> Synthetic test-data generation                           | <input type="checkbox"/> SAST / SCA vulnerability triage and prioritization   |
| <input type="checkbox"/> Secure-code remediation recommendations or pull requests | <input type="checkbox"/> Release evidence and readiness summaries             |

### 31. What minimum net reduction in targeted human effort would justify a second phase?

Select one. "Net" deducts review, correction, and control effort.

### 32. Beyond the effort threshold, which outcomes would justify further investment?

Select up to two.

- |                                                                               |                                                                                     |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <input type="checkbox"/> Material reduction in regression duration            | <input type="checkbox"/> Material reduction in failure-triage time                  |
| <input type="checkbox"/> Material reduction in automation-maintenance effort  | <input type="checkbox"/> No deterioration in escaped defects or change-failure rate |
| <input type="checkbox"/> Material reduction in vulnerability-remediation time | <input type="checkbox"/> Demonstrated contractor reduction or hiring avoidance      |
| <input type="checkbox"/> Credible payback within 12-18 months                 |                                                                                     |

### 33. What conditions should stop or prevent expansion of the pilot?

Select all that apply.

- |                                                                                 |                                                                             |
|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| <input type="checkbox"/> Insufficient measurable savings                        | <input type="checkbox"/> High implementation or integration cost            |
| <input type="checkbox"/> High ongoing license or model-consumption cost         | <input type="checkbox"/> Poor accuracy or excessive human rework            |
| <input type="checkbox"/> Security, privacy, residency, or audit concerns        | <input type="checkbox"/> Inability to integrate with the existing toolchain |
| <input type="checkbox"/> Quality or release-stability deterioration             | <input type="checkbox"/> Low adoption or workflow abandonment               |
| <input type="checkbox"/> No credible mechanism to capture the value financially |                                                                             |

## 6. Pilot direction and executive input

Continued

**34. What is the single most expensive or frustrating part of the current QA or AppSec workflow?**

*Optional.*

**35. What question would the EVP most want the presentation to answer?**

*Optional.*

**36. Is there a representative application, team, or recent release that should be used as an illustrative case, and who is the delivery lead we may contact?**

*Optional.*

Thank you. Responses will be used to prepare a focused executive presentation, an initial savings hypothesis with explicit assumptions, and a conservative phased pilot with a defined cost ceiling and go/no-go decision.